

Loan Default Prediction

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Machine Learning

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1 Introduction

Banks and financial institutions play an important role in providing credit to individuals and businesses. One of the main challenges they face is determining whether a borrower will be able to repay a loan. When borrowers default, financial institutions can experience significant losses, which increases overall lending risk. For this reason, being able to accurately predict loan defaults is essential for managing risk and maintaining sustainable lending practices.

This report explores how machine learning methods can be used to improve the prediction of loan defaults. The main objective is to build a model that identifies borrowers who are more likely to default and to compare the performance of different modeling approaches. In addition to prediction accuracy, this study also examines which borrower characteristics are most closely related to default risk and discusses the trade-offs between model simplicity, accuracy, and real-world applicability.

2 Literature

Predicting loan defaults is an important issue in finance because it affects both lenders and borrowers. When financial institutions can accurately predict which borrowers are more likely to default, they are able to manage credit risk more effectively. This reduces the chance of lending to high-risk individuals while still allowing access to credit for borrowers who are more likely to repay their loans. On the other hand, high default rates can lead to large financial losses, higher interest rates, and increased risk within the financial system. Improving default prediction helps lenders reduce bad debt, lower lending costs, and offer better loan terms to responsible borrowers. At the same time, identifying high-risk borrowers early can allow for interventions such as financial counseling or adjusted repayment plans, which may help prevent more serious financial difficulties.

The need for better loan default prediction has become even more important because consumer debt and delinquency rates in Canada have been rising. By the end of 2024, total consumer debt reached 2.56 trillion dollars, representing a 4.6% increase from the previous year, while non-bank auto loans increased by 11.7% (Equifax Canada, 2025). Mortgage delinquencies also increased, particularly in Ontario, where missed mortgage payments in the fourth quarter of 2024 were nearly three times higher than in 2022. Significant increases were also observed in British Columbia and Quebec. Credit card debt also grew during this period, reaching a seasonal high. Younger and lower-income Canadians were among the groups facing the most financial pressure, many of whom struggle to keep up with payments on credit cards, auto loans, and lines of credit (Equifax Canada, 2025). Together, these patterns highlight the growing financial stress faced by households and emphasize the need for more accurate and reliable loan default prediction tools.

2.1 Machine Learning in Loan Default Prediction

The use of machine learning in credit risk assessment is based on earlier statistical methods that were designed to predict binary outcomes. One of the most common used approaches is logistic regression because is simple, easy to interpret, and produces probabilities that can be used for decision making. This method was originally introduced by Berkson (1944) and later formalized in economics by McFadden (1984). However, logistic regression assumes a linear relationship between the predictors and the probability of default, which can limit its ability to capture more complex patterns.

To overcome these limitations, more advanced machine learning methods were developed and used. Random Forest, introduced by Breiman (2001), combines many decision trees and averages their predictions to improve overall accuracy. This helps reduces overfitting and allows the model to capture more complex relationships in the data. Random Forest is commonly used in financial applications because it can handle both numerical and categorical variables (James et al., 2021). One drawback of this method is that training many trees can be computationally demanding, especially when working with large datasets.

Another commonly used and more advanced method is XGBoost, developed by Chen and Guestrin (2016). Unlike Random Forest, XGBoost builds trees sequentially, with each new tree trying to correct the mistakes of the previous ones. XGBoost also includes regularization techniques to reduce overfitting and is often very effective for structured datasets such as loan and borrower information. This report starts with logistic regression as a baseline and moves towards more advanced models such as Random Forest and XGBoost. It also highlights the trade-off between simplicity, computational cost, interpretability, and predictive performance.

3 Data

The data used in this study comes from Coursera's **Data Science Coding Challenge: Loan Default Prediction** and was obtained through Kaggle (Coursera, n.d.). The dataset contains 255,347 observations and includes 18 variables that describe borrower characteristics, financial information, and loan details. The goal of this dataset is to predict whether a borrower will default on a loan. Although the official competition page does not publicly report leaderboard results or winning submissions, community discussions suggest that strong results are often achieved using machine learning approaches such as Random Forest and XGBoost. This makes the dataset useful for comparing different models and seeing whether more advanced methods can improve prediction accuracy.

The dataset includes both numerical and categorical variables, making it suitable for a range of machine learning methods. The variables cover areas such as demographic characteristics, employment status, loan purpose, credit-related information, and other factors commonly considered in lending decisions. To evaluate the models fairly, 20% of the data was set aside as a holdout test set. This portion of the data was not used during model training or hyperparameter tuning and was only used at the very end for final model evaluation. Using a holdout sample gives a better idea of how the models would perform on new borrowers in a real-world setting. The remaining 80% of the data was split again into a training set and a test set. The training set was used to fit the models, while the test set was used to compare models and tune the best one.

3.1 Variables Considered

The dataset includes the following variables:

- **LoanID:** Unique identifier for each loan.
- **Age:** Borrower's age.
- **Income:** Borrower's annual income.
- **LoanAmount:** Total amount of money borrowed.
- **CreditScore:** Borrower's credit score.
- **MonthsEmployed:** Number of months the borrower has been employed.
- **NumCreditLines:** Number of active credit lines held by the borrower.
- **InterestRate:** Interest rate applied to the loan.
- **LoanTerm:** Duration of the loan in months.
- **DTIRatio:** Debt-to-income ratio, measuring the borrower's total debt relative to their income.
- **Education:** Borrower's highest level of education (PhD, Master's, Bachelor's, High School).
- **EmploymentType:** Borrower's employment status (Full-time, Part-time, Self-employed, Unemployed).
- **MaritalStatus:** Borrower's marital status (Single, Married, Divorced).
- **HasMortgage:** Whether the borrower has an active mortgage.
- **HasDependents:** Whether the borrower has dependents.
- **LoanPurpose:** Reason for taking the loan (Home, Auto, Education, Business, Other).
- **HasCoSigner:** Whether the loan has a co-signer.
- **Default:** Target variable showing whether the borrower defaulted (1 = Yes, 0 = No).

3.2 Exploratory Data Analysis

Before building the models, an exploratory data analysis was done to better understand the structure of the dataset and the relationship between key variables and loan default. Since loan default is typically a rare event in real-world credit markets, the first step is to examine the overall class balance. Figure 1 shows that approximately 88% of borrowers did not default, while only 12% defaulted. This imbalance is important because the model could achieve high accuracy simply by predicting “No Default” for most borrowers. For this reason, metrics such as AUC, precision, and recall are particularly relevant in this context.

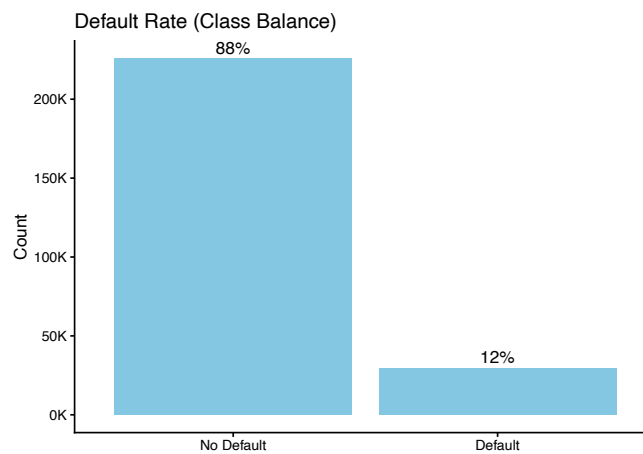


Figure 1

Credit score is one of the most common measures of borrower risk, so it was expected to show a relationship with default. The boxplots in Figure 2 shows defaulters tend to have low credit scores on average but there is considerable overlap between the two groups. This suggests that credit score is relevant but not sufficient on its own to perfectly separate defaulters from non-defaulters.

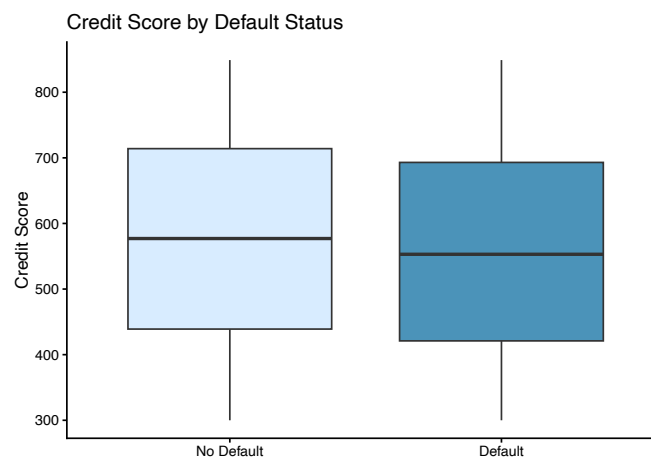


Figure 2

Income is another factor commonly considered in credit risk assessment because it reflects a borrower's ability to repay debt. Figure 3 shows that borrowers who defaulted tend to have lower income on average than borrowers who did not default. The distribution of defaulters is shifted slightly downward compared to non-defaulters, suggesting that income may be an important predictor in the model.

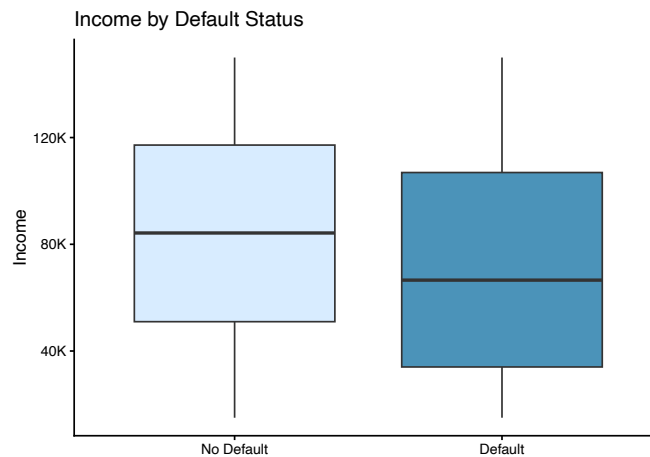


Figure 3

Higher interest rates are usually given to riskier borrowers and at the same time, higher rates make loans more expensive and harder to repay, which may increase default risk. Figure 4 supports this idea since borrowers who defaulted tended to have higher interest rates on average. This suggests that interest rate may capture both borrower risk and financial pressure. The difference between the two groups also looks more clearer than for credit score, which may mean that interest rate could be a strong predictor of default in the model.

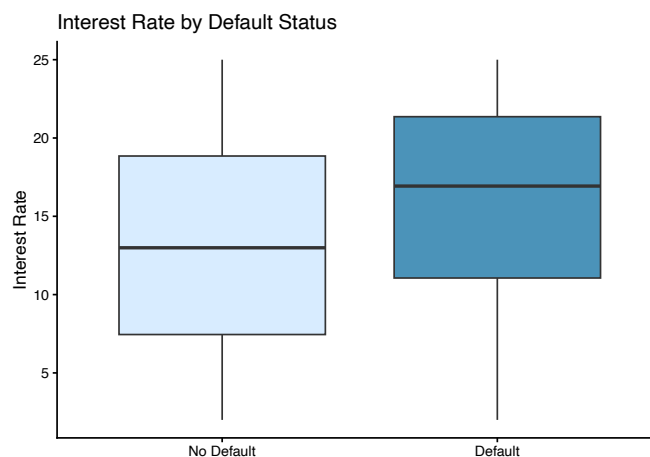


Figure 4

The exploratory analysis shows that borrowers who defaulted tend to have lower incomes, lower credit scores, and higher interest rates. At the same time, there is still some overlap between the two groups in all three variables, especially credit score. This means no single variable can perfectly predict default by itself, which reinforces the need for machine learning models capable of capturing nonlinear relationships and interactions between variables.

3.3 Data Preparation

After exploring the dataset, several steps were performed to prepare the data for the models. First, 20% of the full dataset was randomly selected and set aside as a holdout sample. This holdout sample was not used during training or tuning and was only used for the final evaluation. This shows how the model would perform on new borrowers in a real-world setting. The remaining 80% of the full dataset was then split into training and test data. The training data (80%) was used to fit the models, while the test data (20%) was used to compare the model performance and tune the best model. The proportion of default cases was checked after each split to make sure that the share of defaults stayed similar across the full dataset, training data, test data, and holdout sample. Also, a fixed random seed was used before each split to make sure that the results could be reproduced.

3.3.1 Data Preparation for Machine Learning Models

Different machine learning models require the data to be structured in slightly different ways. The dataset was prepared separately for each model while maintaining a consistent set of variables across all models. First, LoanID was removed because it only serves as an identifier and does not contain useful information for prediction. Variables with Yes/No answers, such as HasMortgage, HasDependents, and HasCoSigner, were converted into binary values of 1 and 0. Categorical variables such as Education, EmploymentType, MaritalStatus, and LoanPurpose were converted into factor variables.

For Logistic Regression, the cleaned dataset could be used directly because the model can handle factor variables automatically. For Random Forest, the process was similar, but the target variable Default was converted into a factor variable because Random Forest treats the outcome as a category. On the other hand, all variables need to be numeric for XGBoost. Because of this, categorical variables were converted into dummy variables, where each category becomes its own binary column. Preparing the data in this way made sure that each model received the format it needed while still using the same underlying information. This allows the performance of the different models to be compared more fairly.

4 Model Selection

The exploratory analysis showed that loan default is a relatively rare event in the dataset, representing only about 12% of all observations. It also showed that variables such as credit score, income, and interest rate are related to default, but none of them can predict default on their own. This suggests that models which combine information from several variables and capture more complex patterns may perform better. Based on these observations, three models were selected for comparison: Logistic Regression, Random Forest, and XGBoost. These models represent different levels of complexity and allow for a comparison between a simple baseline model and more advanced machine learning methods.

Other models, such as k-nearest neighbour (KNN) and single decision trees, were considered but not included in the final comparison. KNN can become unreliable when there are many variables and can also be slow with large datasets. On the other hand, single decision trees are easy to understand but often overfit the training data.

Logistic Regression was selected as the baseline because it is commonly used in credit risk analysis and is easy to interpret. It estimates the probability of default and helps show how different variables affect the probability. However, it assumes a linear relationship between the predictors and the outcome, which may limit its ability to capture more complex patterns in the data. Random Forest was included because it combines many decision trees instead of relying on a single tree. This allows the model to capture nonlinear relationships and interactions between variables. Using many trees also helps reduce overfitting and often improves predictive accuracy. XGBoost was selected because it is often one of the best-performing models for structured datasets such as loan data. Unlike Random Forest, XGBoost builds trees sequentially, with each new tree trying to correct the mistakes of the previous one. This allows the model to capture more complex patterns in the data.

All three models were trained using the training data and evaluated using the test data. Mean Squared Error (MSE) was used as the main measure of model performance because all models generate predicted probabilities of default. A lower MSE means that the predicted probabilities are closer to the actual outcomes. After comparing the three models, the model with the lowest MSE was tuned further by testing different combinations of parameters. MSE was also used during tuning to find the best-performing version of the model. Other measures such as AUC, accuracy, precision, and recall were also calculated to give a more complete picture of the performance.

5 Logit Model

Logistic Regression, also known as logit model, is one of the most used methods for predicting outcomes that have only two possible values, such as default or no default. Instead of directly predicting a “yes” or “no” outcome, the model estimates the probability that a borrower will default based on variables such as income, credit score, employment history, and loan characteristics. The model works by combining the predictor variables a single value using a linear equation. This value is then transformed into a probability between 0 and 1 using the logistic function. A probability closer to 1 means a borrower is more likely to default, while a probability closer to 0 suggest a lower risk of default. One reason Logistic Regression is commonly used in finance is because it is simple and easy to interpret. It allows researchers to see how different variables are related to the probability of default. Because of this, Logistic Regression is used as a baseline model before moving to more advanced machine learning methods in this report.

5.1 Training the Logit Model

The data was prepared as described in Section 3.3.1 and the logistic regression model was trained using the training dataset. The model was estimated in R using the `glm()` function with a logit link. This allows the model to estimate the probability that a borrower will default based on the variables included in the dataset. After the model was trained, predicted probabilities of default were generated for the test dataset. These probabilities were then compared to the actual outcomes to evaluate how well the model performed on new observations that were not used during training.

5.2 Model Performance

Model performance of the logistic regression model was evaluated using Mean Squared Error (MSE), which measures how close the predicted probabilities are to the real outcomes. Lower values of MSE indicate better predictions because the model’s estimated probabilities are closer to the true values. The logistic regression model produced an MSE of 0.09267 on the test data. This result was used as a baseline when comparing the performance of Random Forest and XGBoost.

5.3 Variable Importance

Logistic regression can also show which variables have the strongest relationship with loan default. Figure 5 shows the ten most important variables in the model based on the statistical importance of their coefficients. Variables with larger values in the plot have a stronger influence on the model’s predictions. As shown in Figure 5, age appears to be the most important variable

in the model. This suggests that differences in borrower age may be related to different levels of default risk. Interest rate is also highly important, which makes sense since higher interest rates increase the cost of borrowing and may make loans harder to repay. Other important variables include months employed, income, and loan amount. These variables all give information about the borrower's financial stability and ability to repay the loan.

Some personal characteristics also appear in the importance plot. For example, being unemployed or working part-time, as well as whether the borrower has a co-signer or dependents, also seem to affect the probability of default. Interestingly, credit score appears near the bottom of the ranking, even though it is usually considered one of the main indicators of credit risk. One possible explanation is that other variables, such as income, employment stability, or loan conditions, may already capture part of the information that credit score represents. Because of this, the model may rely more on those variables when predicting default.

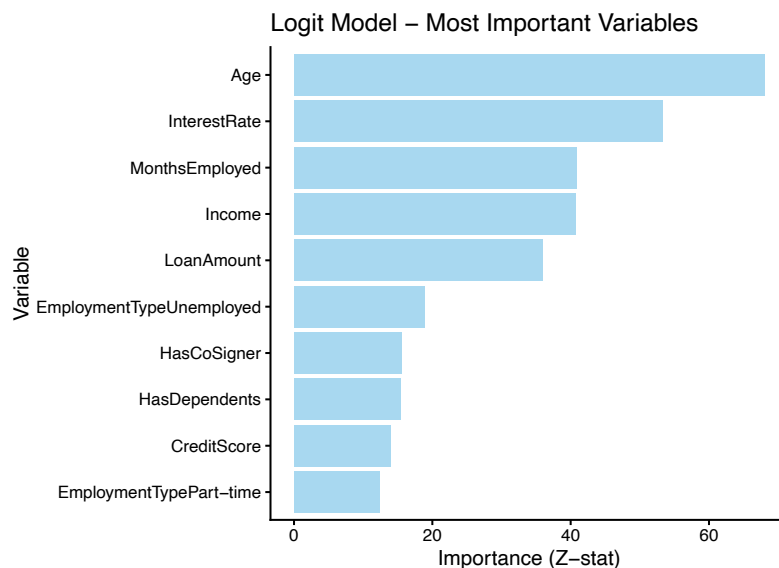


Figure 5

6 Random Forest

Before introducing the Random Forest model, it is useful to first understand how classification trees work. A classification tree makes predictions by splitting the data into smaller groups based on different variables. At each step, the model asks a simple question about one of the variables, such as whether a borrower's income is above a certain level or whether their credit score is above a certain value. Based on the answer, the data is split again until the model reaches a final prediction. Classification trees are easy to understand because they work like a flowchart of decisions. However, a single tree can easily overfit the training data, which means it may learn patterns that only exist in the training data and do not apply well to new borrowers.

As a result, predictions from a single tree can sometimes be unstable and sensitive to small changes in the data. Random Forest was created to solve this problem. Instead of using only one tree, Random Forest builds many trees and combines their predictions. Each tree is trained using a random sample of the data and a random sample of the variables. By averaging the predictions from many trees, the model usually becomes more stable and less likely to overfit (Breiman, 2001). Random Forest is useful for datasets like this one because it can capture more complex relationships between variables than Logistic Regression. For this reason, it was included to see whether a more flexible model could improve prediction accuracy.

6.1 Training the Random Forest Model

The data was prepared as described in Section 3.3.1 and the Random Forest model was trained using the training data with the `randomForest()` function from the `randomForest` package in R. Two main parameters were selected when training the model: **`ntree`** and **`mtry`**. The parameter **`ntree`** controls how many trees are built before combining their predictions. Many trees usually make the model more stable. A common guideline is to use around ten times the number of variables (Boehmke & Greenwell, 2019). Since this dataset has 17 variables, the model used 200 trees.

On the other hand, the parameter **`mtry`** controls how many variables are considered at each split in the tree. For classification problems, a common rule is to use a value close to the square root of the total number of variables (Boehmke & Greenwell, 2019). Since the dataset has 17 variables, **`mtry`** was set to 5. These parameters provide a reasonable starting point to evaluate the performance of the model before testing more advanced tuning methods.

6.2 Model Performance

The Random Forest model produced an MSE of 0.09384 on the test data, which is slightly higher than the Logistic Regression model (MSE = 0.09267). This means that Random Forest did not improve prediction accuracy in this case. One possible reason is that the relationships in the dataset may not be very complex, so the additional flexibility of Random Forest did not provide much benefit.

Another reason could be that Random Forest often needs more tuning of their parameters to reach its best performance. The values used for `ntree` and `mtry` in this analysis were chosen as good starting points, but different parameter combinations may lead to better results. This result shows that more complex models do not always guarantee better predictions, in some cases, simpler models can perform just as well depending on the structure of the data.

6.3 Variable Importance

Random Forest can also show which variables are most important for making predictions. Figure 6 shows the ten most important variables in the Random Forest model. Income appears to be the most important variable in the model. This suggests that a borrower's financial situation plays a major role in whether they default on a loan. Borrowers with higher income levels may be better able to keep up with payments. Interest rate and loan amount are also among the most important variables. This suggests that size and cost of the loan matter a lot when predicting default. Larger loans and higher interest rates increase the financial burden on the borrower, which can make default more likely.

Also, credit score is more important in the Random Forest model than in the Logistic Regression model. This may suggest that Random Forest is better at finding more complex patterns involving credit score. For example, credit score may become more important only for certain income levels or loan amounts.

Other variables such as months employed, debt-to-income ratio (DTIRatio), and age are also important because they provide more information about financial stability. Borrowers who have been employed for longer periods or have lower debt compared to their income may be in a better position to repay their loans. These results suggest that Random Forest mainly predicts default by looking at the borrower's financial situation rather than focusing on only one variable.

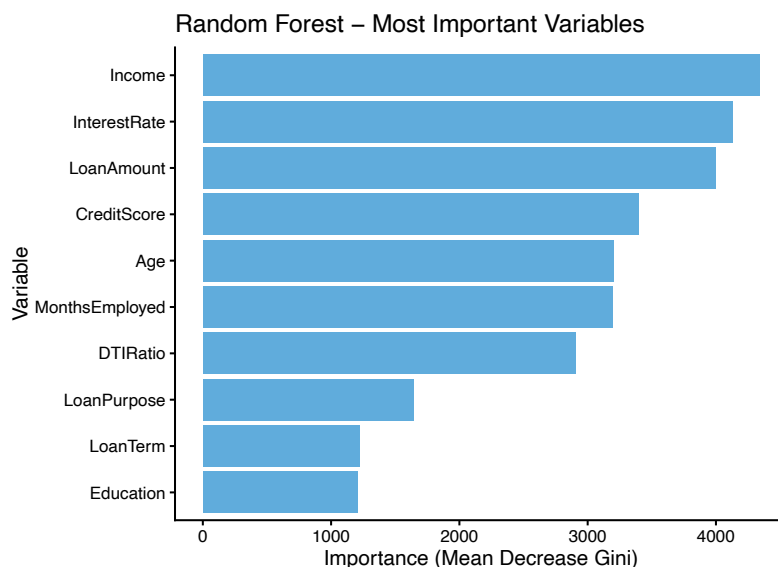


Figure 6

7 XGBoost

The idea behind XGBoost comes from gradient boosting methods. Early boosting models, such as AdaBoost were first introduced by Yoav Freund and Robert Schapire in 1996. XGBoost was later developed by Tianqi Chen and Carlos Guestrin in 2016 as a faster and more efficient version of gradient boosting. XGBoost is based on decision trees, but unlike Random Forest, it build trees one at a time where each new tree focuses on correcting the mistakes made by the previous tree. This process is called boosting. Because the model keeps improving step by step, XGBoost is often able to find more complex patterns in the data than simpler models. It also includes built-in methods to reduce overfitting, which helps the model to perform better on new observations. For these reasons, XGBoost is commonly used for structured datasets such as loan and borrower information.

7.1 Training the XGBoost Model

The data was initially prepared as described in Section 3.3.1, but since XGBoost requires all variables to be numeric, categorical variables such as education, type of employment, purpose of the loan, and marital status were converted into dummy variables. This means that each category was turned into its own binary column. Variables with Yes/No responses, such as having a mortgage, having dependents, and having co-signer were converted into values of 1 and 0. These steps made the dataset fully numeric and compatible with the XGBoost model. The model was trained using the `xgboost()` function in R with several parameters controlling model complexity.

The maximum tree depth was set to 3 because shallow trees are often recommended as a good starting point for boosting models since they can capture important relationships while still reducing the risk of overfitting. The number of boosting rounds was set to 100 and the learning rate (**eta**) was set to 0.1. This is a common baseline value because it provides a balance between learning speed and model stability. Finally, the regularization parameter (**lambda**) was set to 1. This parameter makes the model more conservative and helps reduce overfitting by penalizing very large weights in the model (Boehmke & Greenwell, 2019, p. 227 - 228). These parameter choices provide a reasonable baseline setting before performing more extensive hyperparameter tuning.

7.1.1 Understanding `max_depth`, `nrounds`, and `lambda` in XGBoost

Several parameters play an important role in XGBoost:

- **max_depth** controls how complex each decision tree can be.
- **nrounds** controls how many times the model updates itself.
- **eta** (learning rate) controls how quickly the model learns.

- **lambda** helps reduce overfitting by making the model more conservative.

7.2 Model Performance

The XGBoost model achieved an MSE of 0.09222, which is lower than both the logistic regression (MSE = 0.09267) and the Random Forest model (MSE = 0.09384). This suggests that XGBoost was better at finding the relationships between the variables and loan default. Since the model builds trees step by step and corrects previous mistakes, it may be better at capturing patterns that simpler model miss. Because it had the lowest MSE of the three baseline models, XGBoost was selected for further tuning.

7.3 Variable Importance

Looking at variable importance helps explain why the XGBoost model performed better than the other baseline models. The fact that age, interest rate, income, loan amount, and months employed are the most important variables suggests that the model is mostly identifying borrowers who may be under more financial pressure. For example, borrowers with lower income, larger loans, higher interest rates, or shorter employment history may be more likely to have difficulty making payments. The large drop in importance after the top five variables suggests that most of the predictive power of the model comes from a few key factors. The remaining variables still help improve the predictions, but they appear to play a smaller role. One interesting result is that credit score is not among the top variables, even though it is usually seen as one of the main measures of borrower risk. This may be because other variables, such as income, employment history, and interest rate, already capture similar information about a borrower's financial situation. These results suggest that loan default in this data set is influenced more by a borrower's current financial situation and loan conditions than by personal characteristics alone.

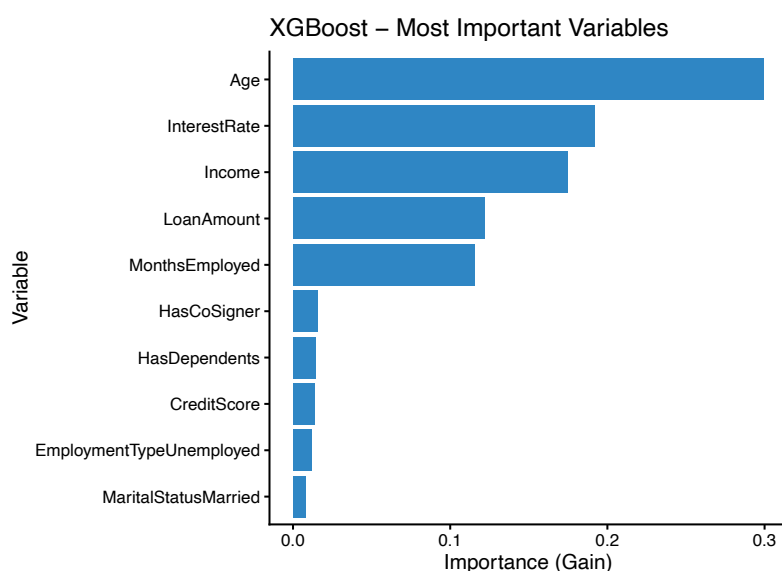


Figure 7

8 Model Tuning

After comparing the baseline models, XGBoost was selected for further tuning because it produced the lowest MSE. The goal of tuning is to test different parameter values and find the combination that gives the best predictions on the test data. A grid search approach was used for this process. This means that several possible values were selected for a few important parameters, and the model was trained repeatedly using every possible combination of those values. In this case, the parameters **max_depth**, **lambda**, and **eta** were tuned. The parameter **nround** handled differently. Instead of testing fixed values, the model was allowed to run up to 1,000 rounds with early stopping. If the model did not improve for 30 rounds in a row, training stopped automatically. This is useful because the best number of rounds depends on the other parameter values being used.

Since many models had to be trained, parallel processing was used to reduce the runtime. Multiple models were trained at the same time using different CPU cores, while leaving two cores unused so the computer would not become overloaded. A progress bar was also included to make it easier to track the tuning process while it was running. One possible improvement would have been to use random search instead of grid search. Random search can sometimes be more efficient because it tests fewer combinations while still exploring a wide range of values. However, grid search was preferred because it is more systematic and easier to interpret. Since only a small number of parameter values were tested, the total number of combinations remained manageable.

8.1 Results

The tuning process found that the best parameter values were **max_depth** = 2, **nrounds** = 385, **lambda** = 1, **eta** = 0.15. These values were used to train the final tuned XGBoost model, and its performance was compared with the baseline of the previous models. The results are shown in Table 1. The tuned XGBoost model achieved the lowest MSE of all four models, with a value of 0.09157. Even though the improvement compared to the baseline XGBoost model was small, it still performed better than Logistic Regression and Random Forest.

Other metrics, such as accuracy, precision, recall, and AUC, were also used to compare the models. Accuracy measures how often the model makes the correct prediction. Precision measures how often the model is correct when it predicts that a borrower will default. Think about high accuracy hitting the bullseye in a target while high precision means all shots are tightly grouped, even if they are far from the center. Recall measures how many of the borrowers who defaulted were correctly identified by the model. AUC (Area Under Curve) measures how well the model can separate borrowers who defaulted from those who do not across many possible probability thresholds. These extra measures are all useful because they focus on different parts of model performance. For example, a model can have high accuracy simply by predicting that most borrowers will not default, since defaults are rare in the dataset. However, this may still lead to poor recall if the model misses many of the borrowers who defaulted.

Model	MSE	Accuracy	Precision	Recall	AUC
Logistic Regression	0.09267	0.88408	0.65164	0.03306	0.51535
Random Forest	0.09384	0.88396	0.57823	0.05301	0.52393
Untuned XGBoost	0.09222	0.88439	0.65052	0.03909	0.51814
Tuned XGBoost	0.09157	0.88530	0.62157	0.06590	0.53028

Table 1

As shown in Table 1, the tuned XGBoost model achieved the highest accuracy, recall, and AUC. This suggests that it was better at identifying high-risk borrowers than the other models. This is important because missing a borrower who will default can be more costly than incorrectly flagging a borrower as risky. Also, the tuned model had slightly lower precision than Logistic Regression and the baseline XGBoost mode. This means it made more default predictions that turned out to be incorrect. However, this trade-off is worth it because the model was able to identify more of the borrowers who defaulted. These results suggest that tuning improved the XGBoost model and made it more useful for identifying more of the borrowers who defaulted.

8.2 Variable Importance

The tuned XGBoost model still relies mostly on the same variables as the baseline version. As shown in Figure 8, age, interest rate, income, loan amount, and months employed remain the most important variables. This shows that the tuning process improved the predictions without changing the main variables driving default risk. One difference is that the tuned model places more importance on financial history variables such as credit score and number of credit lines.

Also, variables such as marital status and dependents become less important. This suggests that the tuned model relies more on information related to a borrower's financial behaviour and existing debt rather than personal characteristics. For example, the number of credit lines may reflect how much debt a borrower is already managing, while credit score provides information about past repayment behaviour. Compared to the baseline XGBoost model shown in Figure 7, Figure 8 shows that the tuned model focuses more on financial variables and borrowing history. This can explain why the tuned model performed better and was better at identifying borrowers who are more likely to default.

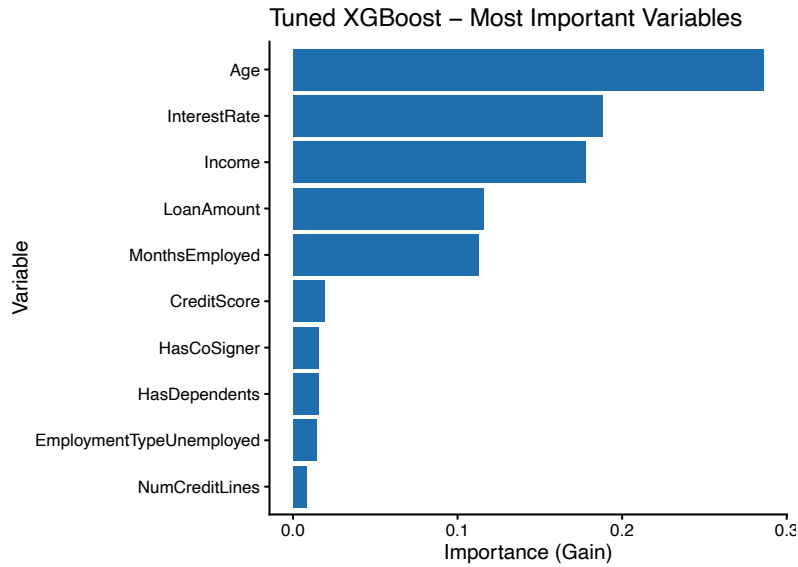


Figure 8

9 Final Model Performance

After selecting the tuned XGBoost model as the best-performing model, all four models were tested one final time using the holdout sample that was separated at the start of the analysis. Since this holdout data was not used during training or tuning, it gives a better idea of how the model would perform on completely new borrowers. The results are shown in Table 2. The tuned XGBoost model again performed best overall. It had the lowest MSE and the highest values for accuracy, recall, and AUC. This shows that the tuning process improved the model and helped it make better predictions than Logistic Regression, Random Forest, and the baseline XGBoost model.

Model	MSE	Accuracy	Precision	Recall	AUC
Tuned XGBoost	0.09087	0.88727	0.63453	0.06880	0.53180
Untuned XGBoost	0.09146	0.88644	0.67282	0.04300	0.52013
Logit Regression	0.09212	0.88551	0.61756	0.03676	0.51689
Random Forest	0.09339	0.88594	0.59849	0.05379	0.52453

Table 2

One of the most important results is that the tuned model had the highest recall among all models. This means the tuned XGBoost model was able to identify more borrowers who defaulted. This is important because missing high-risk borrowers can be more costly than incorrectly flagging some low-risk borrowers. The tuned model did not have the highest precision, which means that some borrowers predicted to default would not default. However, this may still be acceptable because the model was better at identifying true default cases. Comparing all four models on the holdout sample is important because it shows that the tuned XGBoost model did not simply perform well on the training and test data by chance, but it was able to identify patterns that apply to new borrowers. It was still the best-performing model when applied to completely new borrowers. This gives more confidence that the model could be useful in a real lending situation.

9.1 Practical Application for Lenders

The final XGBoost model could be useful for banks and other lenders when deciding whether to approve loan applications. For example, the model could estimate the probability that a borrower will default based on information such as age, income, loan amount, interest rate, employment history, and credit score. Borrowers with a higher predicted probability of default could then be reviewed more carefully before being approved.

Lenders could also use the model to adjust loan conditions. High-risk borrowers could receive smaller loans, higher interest rates, or be required to have a co-signer. The model could also be useful after a loan has already been approved. If a borrower begins to show signs of higher default risk, the lender could identify that borrower earlier and take action before payments are missed. However, the model should not be used on its own. Human judgment is still important because there may be information that is not included in the dataset. Even so, the tuned model could be a useful tool to help lenders make better and more informed decision.

10 Conclusion

The results of this analysis show that machine learning models can be useful for predicting loan default. Although all three baseline models performed reasonably well, the tuned XGBoost model produced the best overall results. It had the lowest MSE and performed best on recall and AUC, which suggests that it was better at identifying borrowers who were more likely to default.

The comparison between models also showed that more complex models do not always lead to better performance. Logistic Regression performed surprisingly well even though it was the simplest model, while Random Forest did not improve the predictions as much as expected. On the other hand, XGBoost performed best because it was able to capture more complex patterns in the data, and the tuning process helped improve it even further.

The variable importance results showed that age, income, interest rate, loan amount, and months employed were some of the most important variables across the models. This suggests that a borrower's financial situation and the conditions of the loan are both important when predicting default risk. Another important result is that the tuned XGBoost model also performed best on the holdout sample. Since this data was not used during training or tuning, this result shows that the model can generalize well to new borrowers instead of only memorizing the training data. In a real lending situation, this model could help banks and other lenders make better decisions. It could be used to identify high-risk borrowers, review applications more carefully, and adjust loan terms based on the predicted risk of default. Even though the model should not replace human judgement, it could still be a useful tool to support more informed and consistent lending decisions.

11 Works Cited

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